

Customer Churn Analysis — Telco Dataset

Pipeline active

Ujjawal2507 / customer-churn-prediction

7,043 records · 19 features · 4 models



 Export

· Aug 2025

CHURN RATE

26.5%

of 7,043 customers

▲ Minority class

BEST AUC-ROC

0.845

XGBoost selected

↑ +2.1% vs baseline

RECALL ACHIEVED

≥ 80%

threshold = 0.32

Optimized

HIGH RISK

31%

flagged customers

Immediate action

MODELS TRAINED

4

GB · RF · LR · XGB

SMOTE balanced

ROC Curve Comparison

XGBGBRF LR

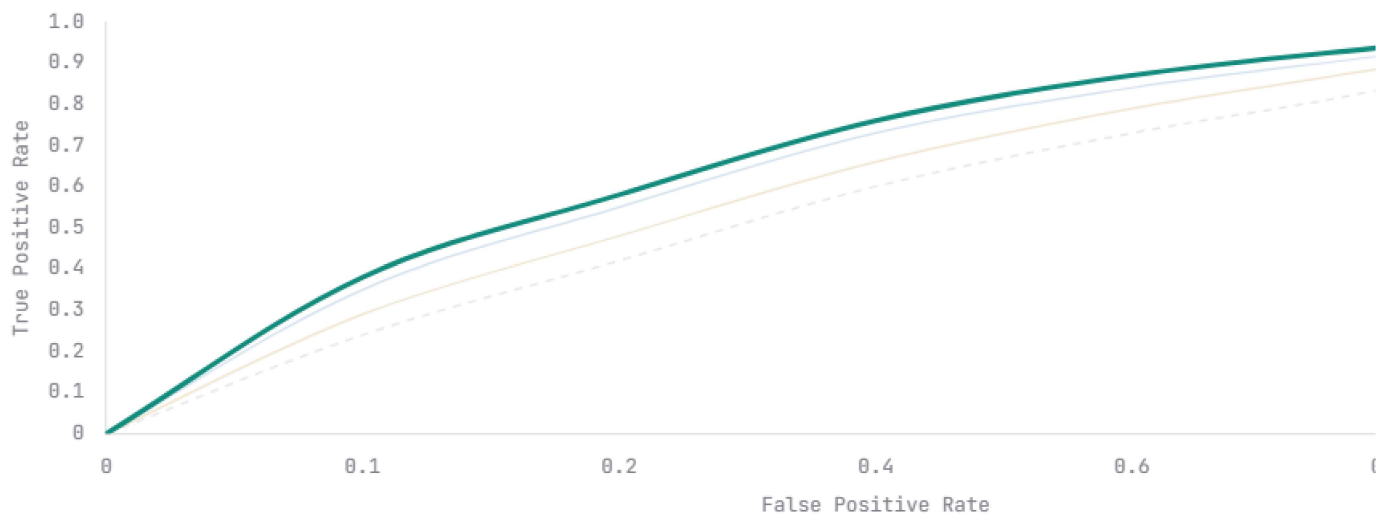
All four models plotted — click a tab to highlight that model's curve against the others.

XGBoost 0.845

Grad. Boosting 0.831

Random Forest 0.814

Logistic Reg. 0.788



Risk Segmentation

Customer distribution across three retention priority tiers.

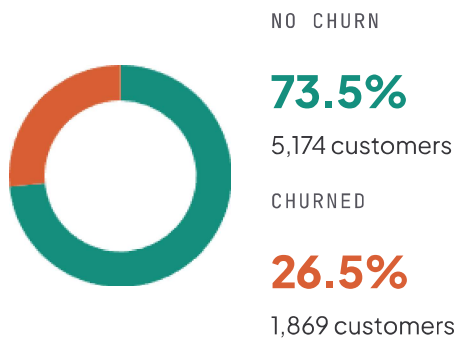
High	31%
Medium	24%
Low	45%

Month-to-month contract holders make up most of the high-risk segment at a 42% churn rate — three times the overall average.

— ChurnLens insight engine

Churn Distribution

Overall class balance in the training dataset.



Feature Importance

Relative contribution of each variable to the best model's predictions.

Contract type	28%
Tenure	22%
Monthly charges	18%
Total charges	14%
Internet service	10%
Payment method	8%

Contract Type Breakdown

Churn rate varies dramatically by commitment tier.

MONTH-TO-MONTH

42%

Highest risk. No lock-in means any friction event triggers exit. Immediate retention priority.

ONE YEAR

11%

Moderate risk. Suitable for upsell and service bundling during renewal windows.

TWO YEAR

3%

Benchmark cohort. Use patterns here to define the ideal customer profile.

Monthly Charges vs Churn

Churned customers cluster in the \$65–\$85 band, often without tech-support add-ons.



ML PIPELINE

End-to-End Pipeline

From raw Telco CSV to scored customer risk output — five reproducible stages.

STEP 01

EDA

Churn distribution, numeric histograms, categorical churn rates, correlation heatmap

✓ `eda.py`

STEP 02

Preprocess

Encode categoricals, scale numerics, handle missing TotalCharges values

✓ `churn_pipeline.py`

STEP 03

Balance + Train

SMOTE · `class_weight='balanced'` · 4 models with 5-fold CV

✓ `best_model.pkl`

STEP 04

Threshold Tune

Precision-recall curve · find threshold for $\geq 80\%$ recall · save to `threshold.txt`

✓ `threshold = 0.32`

STEP 05

Score

predict.py scores new CSV → probability, prediction, risk segment output

→ `predict.py`

INFERENCE OUTPUT SAMPLE

Scored Customer Sample

predictions.csv

Output of predict.py — probability, binary prediction, and risk bucket per customer.

CUSTOMER ID	CONTRACT	MONTHLY	CHURN PROB.	PREDICTION	SEGMENT
C-0041	Month-to-month	₹7,250	0.871	1	HIGH
C-0103	One year	₹4,890	0.541	1	MEDIUM
C-0219	Two year	₹2,940	0.090	0	LOW
C-0316	Month-to-month	₹6,580	0.764	1	HIGH
C-0447	One year	₹3,860	0.284	0	LOW

Recall Improvement Strategy

Five techniques applied to maximize churn detection at an acceptable precision tradeoff.

1 Lower classification threshold

Default 0.5 → 0.32. Catch more churners without retraining the model.

+5-10% recall gain

2 SMOTE oversampling

Synthetic minority oversampling on training data to balance the 73 / 27 split.

+5-8% recall gain

3 class_weight = 'balanced'

Penalizes misclassifying the minority class more heavily during training.

+5-8% recall gain

4 XGBoost scale_pos_weight

Auto-calculated ratio ≈ 2.77 passed to XGBoost to handle imbalance natively.

+3-6% recall gain

5 GridSearchCV with scoring = 'recall'

Hyperparameter tuning that directly optimizes recall instead of accuracy.

+2-5% recall gain

ChurnLens · Telco Customer Churn · Kaggle/blastchar ·
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churn_pipeline.py · eda.py · predict.py ·
dashboard.html